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## **Title : JMeter Performance Testing & CI/CD Integration**

**1) Create the JMeter Test Plan** :Create a JMeter test plan containing five Thread Groups.

### **2) Configure the Thread Groups**

Configure each Thread Group with:

- Number of Users: **10**
- Ramp-Up Period: **5 seconds**
- Loop Count: **3**

**3) Configure Test Scenarios:** Add the required HTTP Request samplers, test data, parameters ,listners and assertions for each application also

**4) Execute the Test in Non-GUI Mode:** Execute the JMeter test using the command line to perform the load test efficiently.

### **5) Analyze Results Using Aggregate Graph/Report**

Open the generated JTL file in JMeter and analyze the performance results using the Aggregate Graph/Report.

### **6) Validate Performance Metrics**

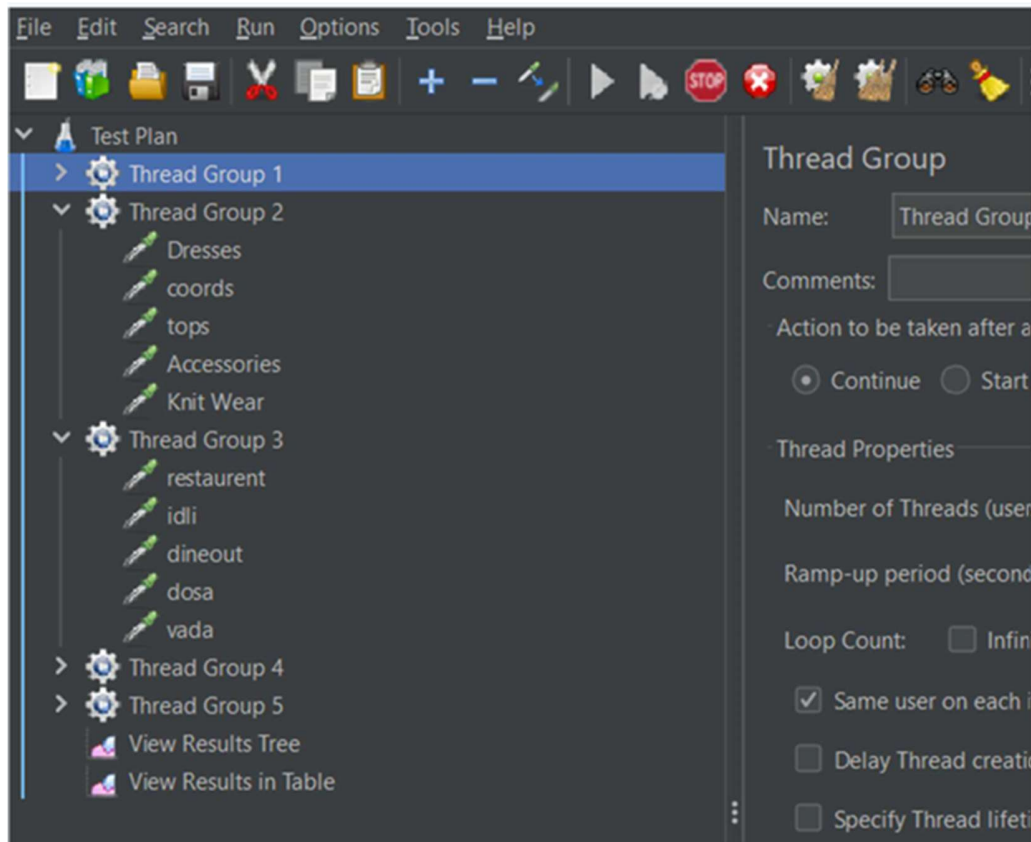
- Total Samples
- Throughput
- Latency
- Median
- Minimum Response Time
- Maximum Response Time
- 90th Percentile
- 95th Percentile
- Error Percentage

### **7) Generate the HTML Dashboard Report and Integrate with jenkins**

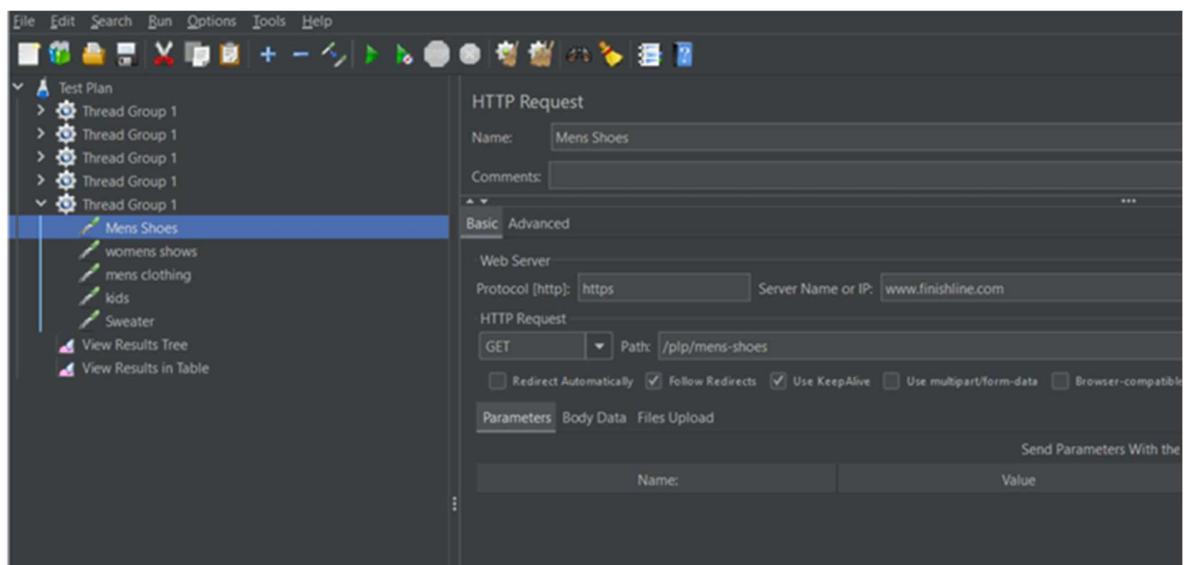
Use the JTL file to generate the JMeter **HTML Dashboard Report** containing graphical performance statistics.

### **8) Validate Jenkins Execution**

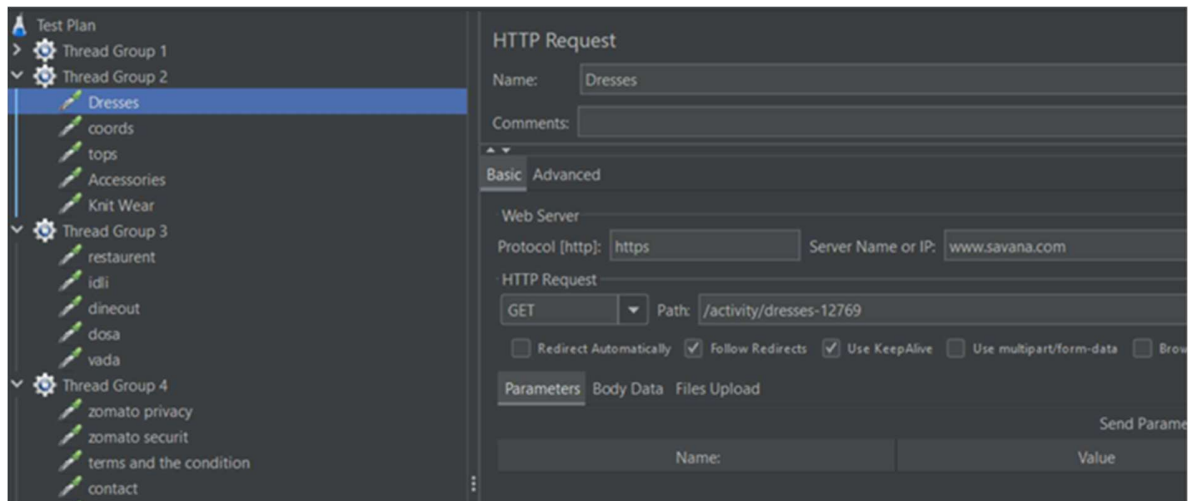
**Step 1:** Create a JMeter test plan containing five Thread Groups



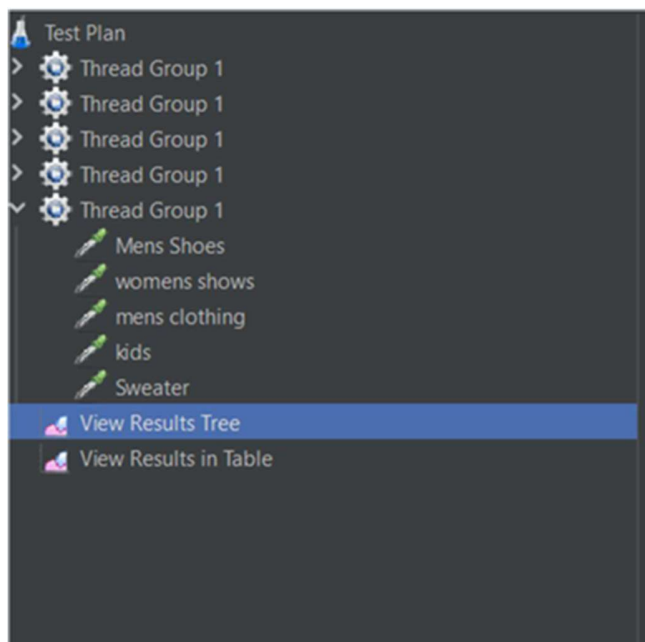
**Step 2:** Configure each Thread Group according to the requirements mentioned



**Step 3:** Add samplers to each thread group using different E-commerce websites



**Step 4:** Add listeners



## Step 5 : Run and validate

View Results in Table

Name: View Results in Table

Comments:

Write results to file / Read from file

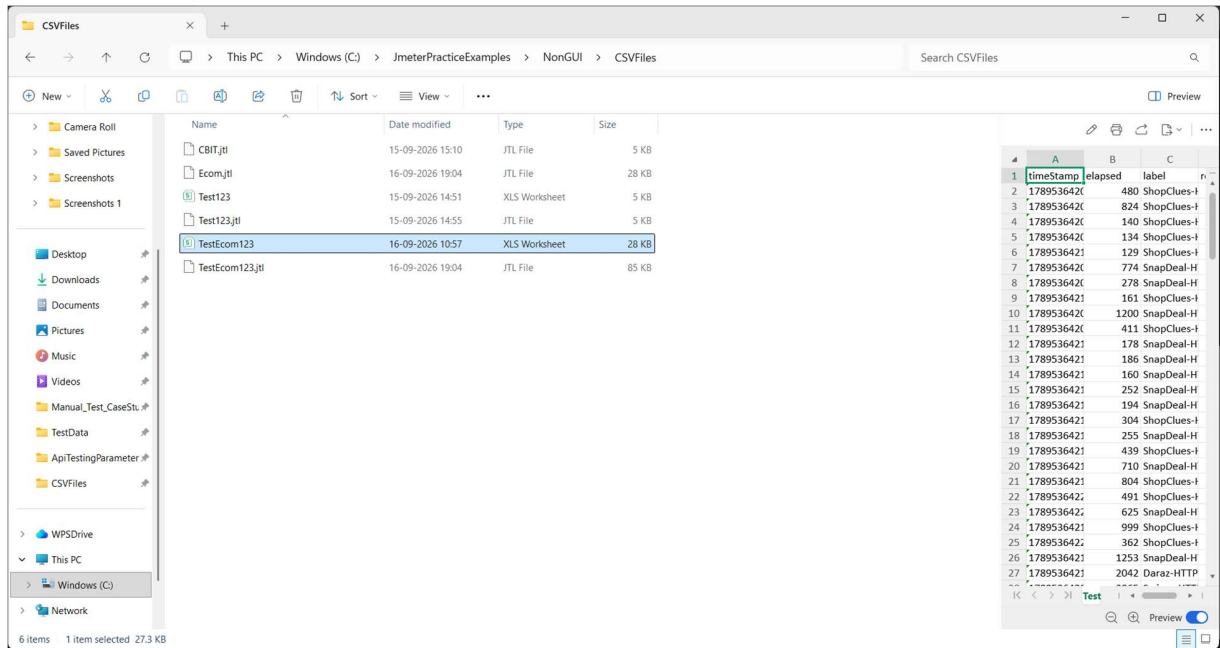
Filename:   Log/Display Only: ☐ Enable

| Sample # | Start Time   | Thread Name          | Label      | Sample Time(ms) | Status | Bytes  | Sent Bytes | ... |
|----------|--------------|----------------------|------------|-----------------|--------|--------|------------|-----|
| 162      | 09:22:39.478 | Thread Group 1 3-... |            | 0               |        | 875    | 0          |     |
| 163      | 09:22:39.102 | Thread Group 1 2-... | coords     | 454             |        | 65221  | 141        |     |
| 164      | 09:22:39.273 | Thread Group 1 3-... | vada       | 310             |        | 92582  | 193        |     |
| 165      | 09:22:37.950 | Thread Group 1 2-... | Dresses    | 1658            |        | 521960 | 141        |     |
| 166      | 09:22:39.248 | Thread Group 1 3-... | vada       | 372             |        | 92604  | 193        |     |
| 167      | 09:22:39.279 | Thread Group 1 2-... | coords     | 375             |        | 67322  | 141        |     |
| 168      | 09:22:39.374 | Thread Group 1 3-... | idli       | 290             |        | 92613  | 207        |     |
| 169      | 09:22:39.479 | Thread Group 1 3-... | vada       | 292             |        | 92597  | 193        |     |
| 170      | 09:22:39.609 | Thread Group 1 2-... | coords     | 187             |        | 67316  | 141        |     |
| 171      | 09:22:38.603 | Thread Group 1 3-... | dineout    | 1198            |        | 318329 | 126        |     |
| 172      | 09:22:39.801 | Thread Group 1 3-... |            | 0               |        | 875    | 0          |     |
| 173      | 09:22:39.408 | Thread Group 1 3-... | idli       | 433             |        | 92605  | 207        |     |
| 174      | 09:22:39.584 | Thread Group 1 3-... | restaurant | 301             |        | 90401  | 130        |     |
| 175      | 09:22:38.863 | Thread Group 1 3-... | dineout    | 1045            |        | 318329 | 126        |     |
| 176      | 09:22:39.908 | Thread Group 1 3-... |            | 0               |        | 875    | 0          |     |
| 177      | 09:22:39.385 | Thread Group 1 3-... | idli       | 566             |        | 92584  | 207        |     |
| 178      | 09:22:39.771 | Thread Group 1 3-... | restaurant | 218             |        | 90400  | 130        |     |
| 179      | 09:22:39.802 | Thread Group 1 3-... | vada       | 241             |        | 92591  | 193        |     |
| 180      | 09:22:39.654 | Thread Group 1 2-... | tops       | 394             |        | 65230  | 157        |     |
| 181      | 09:22:39.796 | Thread Group 1 2-... | tops       | 312             |        | 65234  | 157        |     |
| 182      | 09:22:39.556 | Thread Group 1 2-... | tops       | 560             |        | 67222  | 157        |     |

## Step 6:

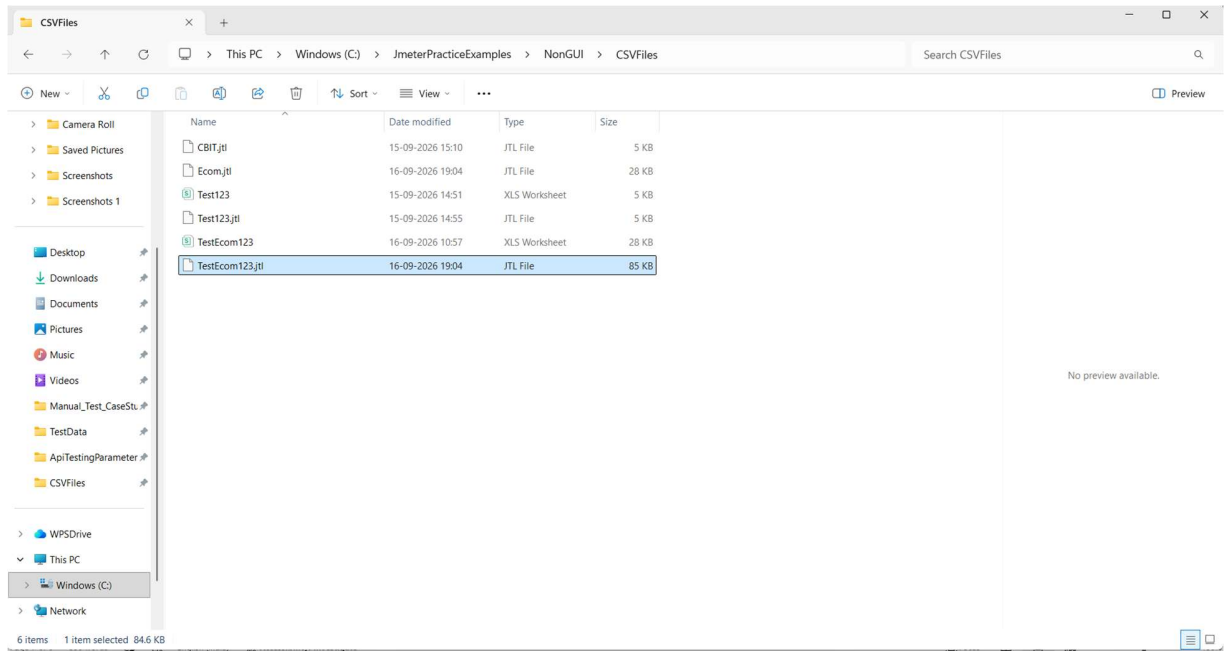
1. Open Command Prompt.
2. Navigate to the JMeter bin folder using the following command:
3. `cd /d "C:\ApacheJmeter5.6.3\apache-jmeter-5.6.3\bin"`
4. Generate the CSV result file by executing the JMeter test plan in non-GUI mode: `jmeter -n -t "C:\JmeterPracticeExamples\NonGUI\NonGUIModeCSVFileEcommerce.jmx" -l "C:\JmeterPracticeExamples\NonGUI\CSVFiles\TestEcom123.csv"`

The test execution results are saved in the specified location as a CSV file.

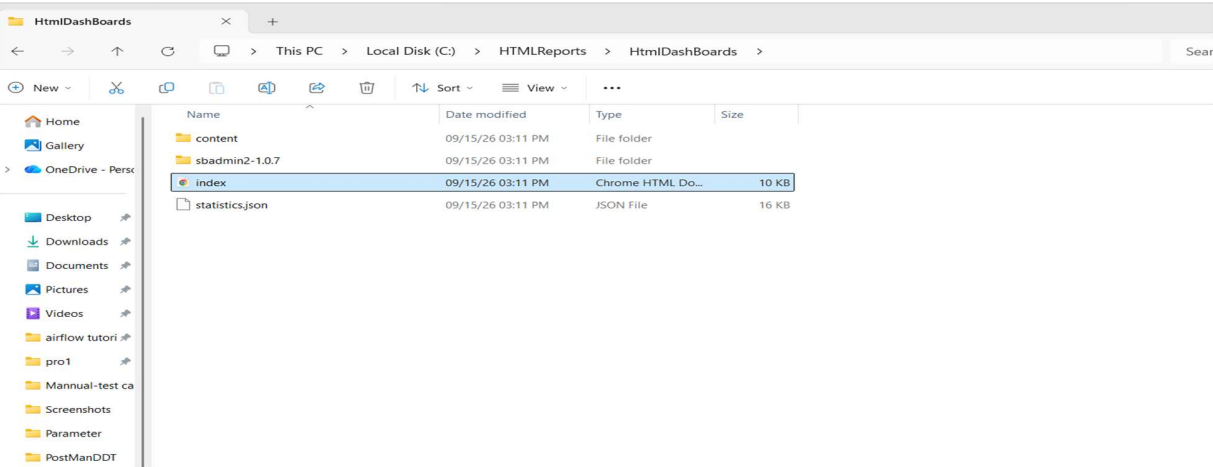


### Step 7:

1. Generate the JTL file to use as the input for the Aggregate Graph: `jmeter -n -t "C:\JmeterPracticeExamples\NonGUI\NonGUIModeCSVFileEcommerce.jmx" -l "C:\JmeterPracticeExamples\NonGUI\CSVFiles\TestEcom123.jtl"`
2. Open the generated JTL file in JMeter and use the Aggregate Graph/Report to analyze the performance results, including Throughput, Latency, Median, Minimum, Maximum, Percentiles, Total Samples, and Error Percentage.

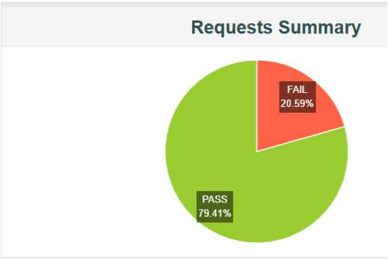


## Step 7: Html reports



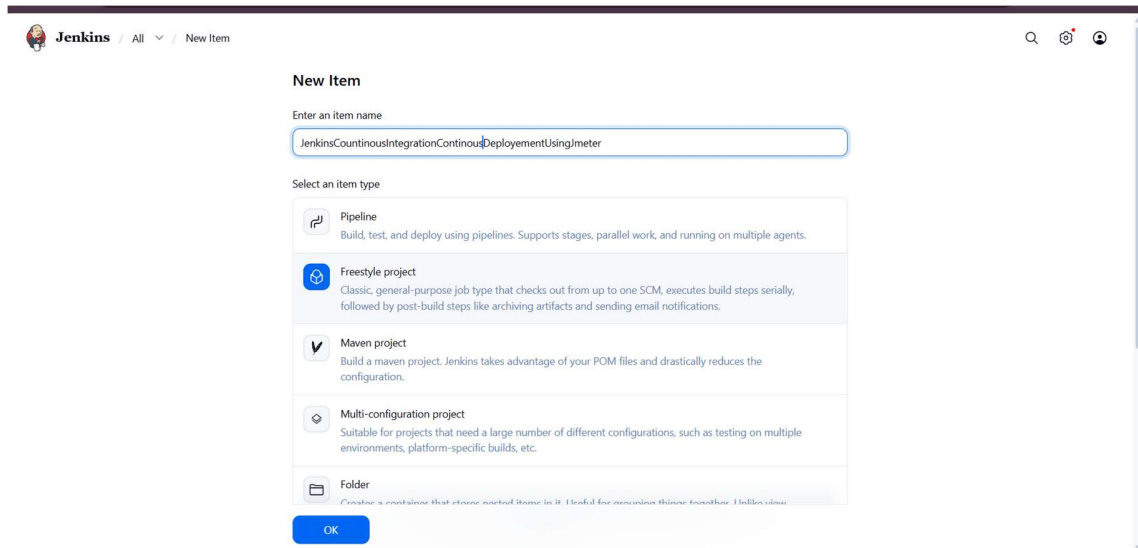
| Test and Report information |                     |
|-----------------------------|---------------------|
| Source file                 | "nongui2.jtl"       |
| Start Time                  | "9/16/26, 10:39 AM" |
| End Time                    | "9/16/26, 10:40 AM" |
| Filter for display          | ""                  |

| APDEX (Application Performance Index) |                          |                           |               |
|---------------------------------------|--------------------------|---------------------------|---------------|
| Apdex                                 | T (Toleration threshold) | F (Frustration threshold) | Label         |
| 0.438                                 | 500 ms                   | 1 sec 500 ms              | Total         |
| 0.000                                 | 500 ms                   | 1 sec 500 ms              |               |
| 0.000                                 | 500 ms                   | 1 sec 500 ms              | Mens Shoes    |
| 0.000                                 | 500 ms                   | 1 sec 500 ms              | mens clothing |
| 0.000                                 | 500 ms                   | 1 sec 500 ms              | womens shows  |
| 0.000                                 | 500 ms                   | 1 sec 500 ms              | kids          |
| 0.200                                 | 500 ms                   | 1 sec 500 ms              | dineout       |

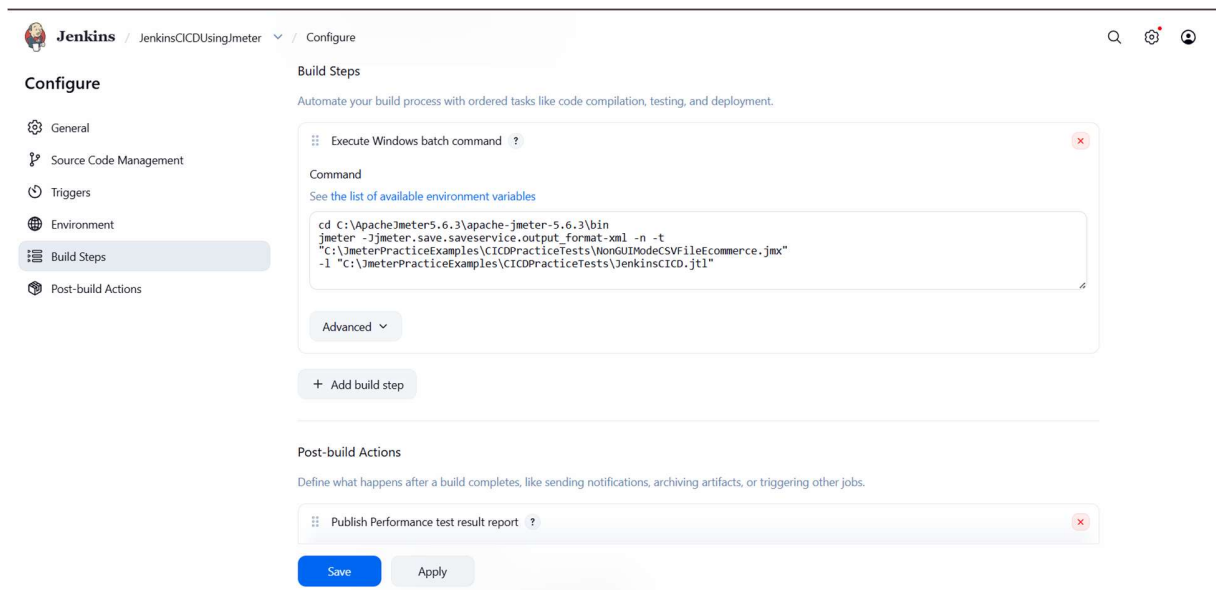


| Statistics    |          |            |         |                     |     |      |         |          |          |          |  |                |                  |
|---------------|----------|------------|---------|---------------------|-----|------|---------|----------|----------|----------|--|----------------|------------------|
| Requests      |          | Executions |         | Response Times (ms) |     |      |         |          |          |          |  | Throughput     |                  |
| Label         | #Samples | FAIL       | Error % | Average             | Min | Max  | Median  | 90th pct | 95th pct | 99th pct |  | Transactions/s | Network (KB/sec) |
| Total         | 340      | 70         | 20.59%  | 1170.94             | 0   | 7968 | 747.00  | 3178.00  | 4623.35  | 6994.43  |  | 14.85          | 2933.74          |
| Accessories   | 30       | 0          | 0.00%   | 1296.60             | 583 | 3526 | 1147.50 | 2378.50  | 3080.50  | 3526.00  |  | 1.56           | 804.01           |
| coords        | 30       | 0          | 0.00%   | 375.57              | 111 | 1585 | 260.00  | 954.80   | 1568.50  | 1585.00  |  | 1.59           | 101.38           |
| dineout       | 30       | 0          | 0.00%   | 2286.83             | 716 | 5437 | 1766.00 | 5020.00  | 5403.45  | 5437.00  |  | 1.37           | 426.78           |
| Dresses       | 30       | 0          | 0.00%   | 1937.20             | 743 | 5254 | 1193.00 | 4584.40  | 5068.65  | 5254.00  |  | 1.52           | 775.47           |
| idil          | 30       | 0          | 0.00%   | 945.97              | 203 | 3182 | 668.50  | 2288.50  | 2802.50  | 3182.00  |  | 1.42           | 128.39           |
| kids          | 10       | 10         | 100.00% | 336.90              | 44  | 1680 | 87.00   | 1564.00  | 1680.00  | 1680.00  |  | 0.90           | 1.46             |
| Knit Wear     | 30       | 0          | 0.00%   | 1415.63             | 615 | 3388 | 1197.00 | 2344.90  | 3001.35  | 3388.00  |  | 1.62           | 808.79           |
| mens clothing | 10       | 10         | 100.00% | 707.80              | 55  | 2929 | 519.00  | 2734.50  | 2929.00  | 2929.00  |  | 0.91           | 1.48             |
| Mens Shoes    | 10       | 10         | 100.00% | 4901.40             | 186 | 7569 | 6477.00 | 7518.80  | 7569.00  | 7569.00  |  | 0.90           | 1.47             |
| restaurant    | 30       | 0          | 0.00%   | 1742.03             | 226 | 7968 | 770.00  | 5260.10  | 6546.25  | 7968.00  |  | 1.40           | 123.39           |
| tops          | 30       | 0          | 0.00%   | 342.10              | 103 | 1367 | 263.00  | 609.40   | 971.00   | 1367.00  |  | 1.59           | 101.32           |
| vada          | 30       | 0          | 0.00%   | 572.13              | 225 | 2469 | 460.00  | 1110.50  | 1875.00  | 2469.00  |  | 1.50           | 135.63           |
| womens shows  | 10       | 10         | 100.00% | 1123.80             | 58  | 3296 | 983.00  | 3134.60  | 3296.00  | 3296.00  |  | 0.91           | 1.48             |
|               | 30       | 30         | 100.00% | 0.00                | 0   | 0    | 0.00    | 0.00     | 0.00     | 0.00     |  | 1.52           | 1.30             |

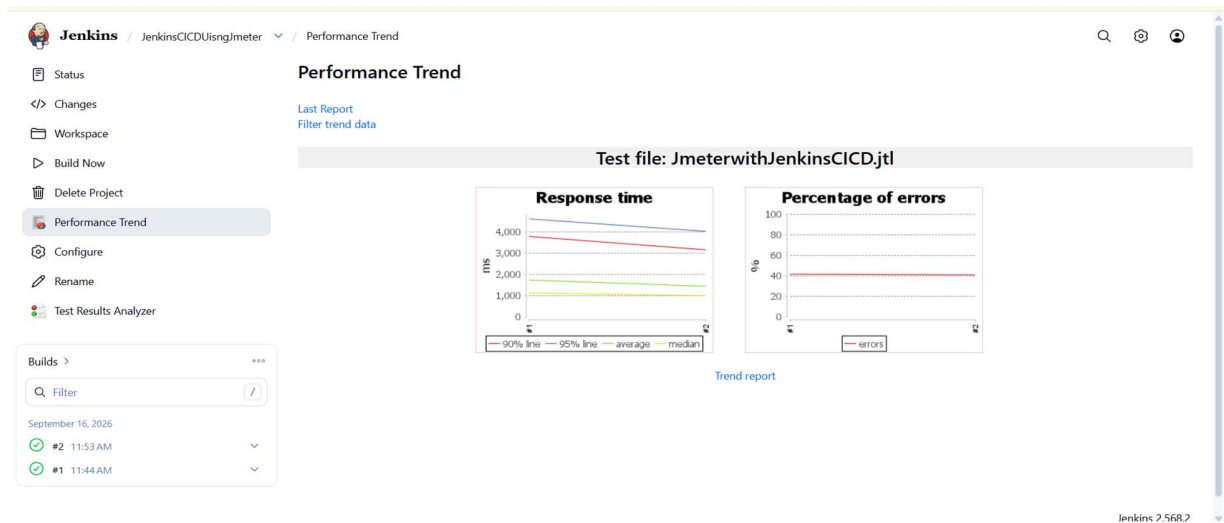
Step 8: Run through Jenkins – create freestyle project



## Step 9 : Configure Jenkins



## Step 10: Analyze the performance Trend



## Step 11: JMeter Aggregate Graph – Performance Test Results

